

Multimodal AAD dataset — analysis iteration log

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Dataset layout — authoritative reference

For every subject the two data stores are parallel copies of the same trials, just partitioned by modality:

`experiment_data/Subject N/`

- Training-1 through Training-5 — 5 familiarisation trials used at the start of recording.
- Eval-1 through Eval-100 — the 100 main trials. Eval-K corresponds to Trial-K in `audio_stimuli_data/trials`.
- Each trial folder holds EEG (`eeg_data.p`, 32ch @ 500 Hz), 2-D gaze (`gaze_data.p`), and `audio_timestamps.json`.

`Video Recordings/Subject N/`

Subfolders numbered 1 through 105:

- Folders 1--5 = the 5 training trials (same ordering as `experiment_data/Training-K`).
- Folders 6--105 = the 100 main trials; folder K corresponds to Eval-(K-5), i.e. Trial-(K-5).
- Each folder holds the Tobii Glasses 3 bundle: `gazedata.gz` (per-eye gaze & pupil), `imudata.gz`, `scenevideo.mp4`, `eventdata.gz`, `recording.g3`, `snap0.jpg`.

The `aad_utils` package enforces this mapping: loaders accept a `kind='main'|'training'` kwarg, and `video_folder_for_trial(k, kind)` translates a logical trial index to the correct Video-Recordings folder number.

Chapter 1

Iteration 1 — initial audit (pre-alignment-fix)

Subjects scanned: 1, 3, 5, 7, 9, 11, 13, 15 (eight subjects); 25 main trials each. These numbers were produced *before* we discovered that the Eval-K folder maps directly to main trial Trial-K (not Trial-(K-5)) — i.e. the EEG was being paired with the wrong audio. Kept here as the baseline for the improvements that followed.

1.1 Summary

Test	Value	Reference
Pooled AAD backward-model accuracy (30 s, 25 trials)	0.422	chance = 0.50; exp
mean $\rho_{\text{att}} - \rho_{\text{una}}$	−0.016	should be positive
Spearman $\rho(\log \text{R/L } \alpha, \text{ attended az})$	−0.122 ($p = 0.12$, $n = 160$)	weak, correct sign
Inter-subject correlation at Cz	0.014	typical 0.05–0.15
Gaze-only 4-class accuracy (chance = 0.25)	0.486 ± 0.274	highly subject-vari
Cerebro-acoustic coherence top channels (δ/θ)	Pz, P3, P7, CP1, T7, T8, O1, O2	canonical auditory

1.2 Per-subject AAD accuracy

Subject	Backward acc.	α -lat. ρ	Gaze-only 4-class
1	0.36	+0.31	0.26
3	0.48	−0.32	0.78
5	0.48	− 0.60	0.45
7	0.44	− 0.58	0.65
9	0.48	−0.06	0.23
11	0.56	+0.10	0.18
13	0.32	+0.29	0.43
15	0.25	−0.26	0.93

1.3 What was suspicious

- Pooled AAD **below chance** is impossible for a correctly-wired pipeline.
- Mean $\rho_{\text{att}} - \rho_{\text{una}}$ is **negative** — the model preferred the unattended envelope.
- ISC of 0.014 is a tenth of the magnitude normal for AAD.

These three signals together triggered the audit that surfaced the trial-indexing bug (Chapter 2).

1.4 What was nevertheless informative

- **Cerebro-acoustic coherence topography** lands on the canonical posterior/temporal pattern for speech tracking, despite the trial-level mis-pairing.
- α -lateralisation in Subjects 5 and 7 shows the textbook strong negative Spearman correlation with attended azimuth — the biomarker survived the bug because it doesn't depend on stimulus identity.
- Gaze is a strong per-subject signal and an obvious multimodal-fusion candidate.

Chapter 2

Trial-alignment bugs (and fixes)

Two related alignment bugs have been found and fixed. Both stemmed from incorrectly inferring the mapping between logical trial indices and on-disk folder structure. Fixing them took AAD pooled accuracy from 0.422 (below chance) to 0.518 — a +9.6 pp jump — and unlocked the multimodal-fusion story.

2.1 Bug A: `experiment_data` folder mapping

2.1.1 Symptoms

- Pooled linear mTRF AAD at **0.422**.
- Mean $\rho_{\text{att}} - \rho_{\text{una}} = -0.016$.
- ISC at Cz = 0.014.

2.1.2 Root cause

I originally assumed that `Eval-1..5` held the 5 training trials and `Eval-6..100` held the first 95 main trials, writing

```
tno = f"Training-{k}" if k <= 5 else f"Trial-{k-5}"
```

for `k` indexing the `Eval` folder. The user clarified: **every subject has 100 main trials in `Eval-1..Eval-100` and 5 training trials in separate `Training-1..Training-5` folders**. The `Training-*` siblings had been missed — so for 95 of 100 main trials, EEG was paired with the previous trial’s audio, randomising the attended/unattended label.

2.1.3 Fix

In `aad_utils/io.py`:

1. `_eval_dir(subject, k)` resolves to `SubjectN/Eval-K` for $k \in [1, 100]$, no shift.
2. New `_training_dir(subject, k)` resolves `Training-K`.
3. New `_trial_dir(subject, k, kind)` and `trial_name(k, kind)` unify both conventions; every loader accepts a `kind='main'|'training'` kwarg.

Every notebook build script and analysis script updated; stale `cache/aad_pairs/` deleted.

2.1.4 Verification (Subject 3)

With corrected alignment and iter-2 improvements:

- Trial-1: $r(\text{Cz}, \text{env}_{\text{att}}) = +0.084$ vs $+0.009$ unattended.
- Trial-2: $r_{\text{att}} = -0.005$ vs $r_{\text{una}} = -0.121$.
- Full 5-fold within-subject backward model (100 trials): **0.520** (was 0.48).

2.2 Bug B: Video Recordings folder offset

2.2.1 Root cause

The Video Recordings/Subject N/ subfolders are numbered 1..105: folders 1..5 are the training trials and folders 6..105 are the 100 main trials (folder $K \rightarrow \text{Eval}-(K-5)$). My original `video_trial_dir(subject, k)` treated k as the folder number directly, so “main trial 1” opened folder 1, which is actually Training-1. Every per-eye gaze / IMU / scene-video analysis for main trials was therefore off by 5 folders.

This bug did *not* affect iter-1/iter-2 AAD numbers (the backward decoder only uses EEG + audio, not Tobii raw gaze). It did corrupt Notebooks 04 (gaze), 07 (gaze-AAD), 08 (cross-modal), 11 (scene video), and 12 (advanced neuro), plus any multimodal-fusion analysis that touched `load_raw_gaze` / `load_raw_imu`.

2.2.2 Fix

New helper in `aad_utils/io.py`:

```
def video_folder_for_trial(k, kind="main"):
    if kind == "training": # 1..5 -> 1..5
        return k
    if kind == "main":     # 1..100 -> 6..105
        return k + 5
```

`video_trial_dir`, `load_raw_gaze`, `load_raw_imu` now accept `kind='main'` (default) and route through this function.

2.2.3 Verification

- `video_folder_for_trial(1, 'main') → 6`.
- `video_folder_for_trial(100, 'main') → 105`.
- `video_folder_for_trial(1, 'training') → 1`.
- Gaze-stream duration in folder $K+5$ matches the audio playback duration in `Eval-K/audio_timestamps.json` within the expected 5–6 s Tobii pre/post-roll.

2.3 Why this matters for the paper

All iter-1 numbers are invalidated. Iter-2 (post-Bug-A) numbers are valid for EEG-only AAD. Any multimodal result that used Tobii streams must be re-run after the Bug-B fix. The corrected pipeline is the basis for iterations 3 and onward.

Chapter 3

Iteration 2 — changes, rationale, SLURM design

Script: `analysis/scripts/aad_v2.py`. Launched 2026-04-18.

3.1 What changed relative to iter-1

#	Change	Why
1	Trial mapping fixed (<code>Eval-K</code> ↔ <code>Trial-K</code>)	the dominant confound in iter-1
2	All 100 main trials per subject	linear TRF is data-hungry
3	ICA with auto-EOG removal (Fp1/Fp2 proxy)	removes blink artefact from 1–9 Hz band
4	Envelope derivative (half-wave rectified) target	Crosse <i>et al.</i> 2016
5	Lags $-100 \dots +400$ ms (was $0 \dots 250$)	captures N1/P2 onset response
6	Ridge α via 80/20 validation split	avoids arbitrary $\alpha=10^3$
7	Window-length sweep (1, 2, 4, 8, 16, 30 s)	classic AAD reporting
8	Correct-only filter (behavioural comprehension)	subjects who paid attention should give cleaner signal
9	Robust bad-channel handling (<code>reference='auto'</code>)	fixes S1 (100 % M1 flat, 100 % M2 sat.) and $\sim 12\%$ o

Runs entirely on CPU. No GPU required.

3.2 SLURM design

Account: PAS2966. Four independent array jobs (16 tasks each = 64 runs):

Job ID	Condition	Flags	Output
4985124	all-trials, ICA+deriv.	<code>CORRECT=0 NO_ICA=0</code>	<code>s*_all_derivative.parquet</code>
4985125	correct-only , ICA+deriv.	<code>CORRECT=1</code>	<code>s*_correct_derivative.parquet</code>
4985126	all-trials, no ICA	<code>NO_ICA=1</code>	<code>s*_all_noica_derivative.parquet</code>
4985127	all-trials, raw env.	<code>TARGET_MODE=raw</code>	<code>s*_all_raw.parquet</code>

Each task: 8 CPUs, 32 GB, 2 h wall.

3.3 Why four conditions

- **all vs correct-only**: does filtering on behavioural correctness help despite the 18% loss of training examples?
- **no-ICA**: how much of the iter-2 improvement comes from ocular cleanup vs. alignment vs. trial count?
- **raw envelope**: isolates the derivative target's contribution.

A companion aggregator (`aggregate_aad_v2.py`) reads every output, computes pooled + per-subject + per-window summaries, and runs paired Wilcoxon tests for the iter-2 results chapter.

Chapter 4

Iteration 2 — results

All 16 subjects $\times$ 4 conditions = 64 SLURM runs, 100 main trials per subject. Trial alignment corrected (Bug A in Chapter 2).

4.1 Headline numbers

Condition	Pooled full-trial acc. (chance = 0.50)	n trials
all · ICA · derivative (winning)	0.518	1599
all · ICA · raw	0.511	1599
all · no-ICA · derivative	0.515	1599
correct · ICA · derivative	0.502	1310

Iter-1 $\rightarrow$ iter-2: 0.422 $\rightarrow$ 0.518 (+9.6 pp).

4.2 Per-subject accuracy (winning condition)

4 of 16 subjects pass 0.55. Bimodal cohort: responsive (S9, S10, S14, S15 at 57–61 %) vs. weak-signal (S1, S4, S7, S12, S13 at 43–48 %).

4.3 Window-length sweep

Flatter than DTU/KULeuven (which climb 55 $\rightarrow$ 75 % over the same range). Two factors: (i) 30s trials are short; (ii) the cocktail plays the same audio from both L/R speakers of a device at different powers, reducing stimulus separability compared to one-talker-per-ear designs.

4.4 Correct-only filter — verdict: does not help

Paired Wilcoxon $p = 0.348$, $n = 16$. Dropped from the primary pipeline. Retained as a diagnostic for a future EEG/behaviour correlation section.

4.5 Ablation verdicts

- **ICA vs no-ICA**: 0.518 vs 0.515. ICA costs $3\times$ runtime for 0.3 pp. Optional.

Subject	AAD accuracy	note
S14	0.610	no-ICA even better (0.65)
S15	0.596	
S9	0.580	
S10	0.570	
S2	0.540	raw envelope better (0.60)
S16	0.540	
S11	0.530	
S3	0.520	
S6	0.520	raw envelope better (0.56)
S5	0.510	
S8	0.490	
S1	0.480	
S4	0.480	known-bad mastoids
S7	0.450	
S12	0.440	
S13	0.430	

Window	mean	std
1 s	0.498	0.010
2 s	0.500	0.011
4 s	0.501	0.020
8 s	0.507	0.032
16 s	0.515	0.048
30 s	0.518	0.054

- **Derivative vs raw:** 0.518 vs 0.511 (+0.7 pp), but subject-heterogeneous (S2, S8 prefer raw).
- α **validation:** iter-2's 80/20 split always picked $\alpha = 10$ (lowest grid). Under-regularised; iter-3 runs proper nested CV over a wider grid.

4.6 Implications for NeurIPS

- Linear TRF alone at ~ 0.52 pooled is weak.
- The per-subject heterogeneity is the story (4 responsive, 5 weak).
- **Multimodal fusion** is the plausibly-NeurIPS-worthy result — gaze alone reached 48 % on 4-class (chance = 0.25) and up to 93 % per-subject. The fusion chapter below makes this concrete.

Condition	mean	Δ vs all
All trials	0.518	
Correct only	0.502	−0.016

Chapter 5

Iteration 3 — plan (revised with iter-2 evidence)

5.1 Decisions carried over from iter-2

1. Drop the correct-only filter from the primary pipeline.
2. Keep ICA off in the primary pipeline (optional flag).
3. Envelope derivative is the default target; per-subject CV may switch to `raw` for S2, S8.
4. Wider ridge α grid with proper nested CV.
5. Subject tiers: responsive (S2, S9, S10, S14, S15, S16) vs. challenging (S1, S4, S7, S12, S13). Report both.

5.2 Iter-3 experiments

5.2.1 A. Linear-decoder improvements (`aad.v3.py`)

Experiment	Description	Target gain
A1 baseline v3	iter-2 winner + nested α -CV + lags $-200 \dots + 500$ ms	+1–2 pp
A2 split $\delta+\theta$	1–4 Hz and 4–8 Hz EEG lags concatenated	+2–4 pp
A3 multi-band envelope	28-band gammatone envelope (mel28) as target	+5–8 pp
A4 CCA baseline	CCA on (EEG lags) $\times$ (mel envelope)	reported
A5 per-subject best-of-3	inner CV picks raw derivative mel	+1–3 pp

5.2.2 B. Multimodal fusion (`aad_fusion.py`) — headline

5.2.3 C. Diagnostics / narrative

1. Per-subject bad-channel rate, α -lat. sign, ISC at Cz, signal quality $\rightarrow$ composite “subject-quality score”; test if it predicts AAD accuracy.
2. Behavioural-accuracy vs. decoding-accuracy scatter.
3. SNR stratification of AAD.

Experiment	Description
B1 Gaze-only (binary AAD)	gaze $\rightarrow$ {attended-speaker group} 2-class classifier
B2 EEG-only (binary AAD)	iter-3 A5 out-of-fold predictions
B3 Late-fusion (prob-avg)	mean of (A5 probs, gaze probs)
B4 Stacked-fusion	LightGBM meta-learner on [A5 oof, gaze oof]
B5 Early-fusion	concat EEG band-power + gaze features $\rightarrow$ LightGBM
B6 Learning curves	accuracy vs. n_{trials} ($\in \{5, 20, 50, 100\}$)

5.3 Success criteria

- Pooled within-subject AAD at 30s ≥ 0.60 (binary) with CI clear of chance.
- Fusion ≥ 0.70 pooled on the responsive subset.
- Single novelty claim — likely the synchronised gaze + video + questionnaires missing from existing AAD datasets.

Chapter 6

Iteration 3 — results (EEG backbones + multimodal fusion)

64 SLURM runs across 16 subjects $\times$ 4 feature backbones, followed by a binary-group-level fusion pipeline (Device-1 vs Device-2) that consumes each backbone’s out-of-fold predictions and combines them with a gaze-only classifier.

6.1 Linear-decoder comparison (Experiment A)

6.1.1 Pooled full-trial accuracy (chance = 0.50)

Backbone	mean	n trials
CCA mel-28 (A4)	0.523	1599
mel-28 ridge (A3)	0.511	1599
broadband ridge (A1)	0.507	1599
split $\delta+\theta$ (A2)	0.492	1599
oracle adaptive (A5: per-subject best)	0.555	1599

CCA on 28-band gammatone envelopes beats every scalar-target ridge variant. Split $\delta+\theta$ actively hurts — likely because doubling the regressor count without doubling the data exacerbates the nested-CV variance of α selection. The oracle “per-subject pick-the-best backbone” hits 0.555, showing that subject-adaptive decoders can buy $\sim 3\text{--}4$ pp over any single uniform choice.

6.1.2 Per-subject winning backbone

Best-of-4 backbone	Subjects
CCA mel-28	S1, S2, S4, S5, S7, S10, S11, S12, S13 (9 subjects)
broadband ridge	S3, S9, S14, S16 (4 subjects)
mel-28 ridge	S6, S15 (2 subjects)
split $\delta+\theta$	S8 (1 subject)

6.1.3 Window sweep (best backbone per feature)

Window (s)	broadband	mel-28	split $\delta+\theta$
1	0.496	0.499	0.500
2	0.499	0.501	0.498
4	0.500	0.505	0.499
8	0.505	0.500	0.502
16	0.510	0.524	0.504
30	0.507	0.510	0.492

All three ridge variants stay near chance for windows ≤ 4 s and only climb modestly at 16–30 s, consistent with the 30 s trial length and the reduced spatial separability of this paradigm.

6.2 Multimodal fusion (Experiment B) — headline

Binary task: does the subject attend the Device-1 speaker pair ($\pm 22.5^\circ$) or the Device-2 pair ($\pm 67.5^\circ$)?

6.2.1 Pooled accuracy across 16 subjects (bootstrap 95 % CI)

Modality / fusion	Pooled accuracy	95 % CI
gaze-only	0.784	[0.697, 0.863]
EEG-only (mel-28 $\rho_{\text{att}} - \rho_{\text{una}}$ sigmoid)	0.491	[0.465, 0.515]
EEG-only (mel-28 LogReg on $\rho_{\text{att}}, \rho_{\text{una}}$)	0.518	[0.493, 0.543]
late fusion (mean prob., mel-28)	0.787	[0.706, 0.863]
stacked fusion (LightGBM, mel-28)	0.782	[0.705, 0.853]
early fusion (concat features, mel-28)	0.797	[0.723, 0.866]

Early-fusion with the mel-28 EEG backbone is the winning configuration at 0.797 pooled. The gain over gaze-only is small (+1.3 pp pooled) because gaze is already near-saturation — it decodes the attended speaker group at ≥ 0.94 for 6 of 16 subjects. The EEG contribution mostly helps the subjects whose gaze drifts.

6.2.2 Per-subject fusion (mel-28 backbone)

6.2.3 Key observations

1. **Gaze carries the signal in this dataset’s spatial paradigm.** Despite no eye-tracker calibration, 6/16 subjects have gaze-based binary AAD ≥ 0.93 . The spatial attention task naturally causes subjects to look toward the attended source.
2. **EEG on its own is at chance for binary group-level AAD.** This is not a bug: the mTRF models extract trial-level correlation differences, not group-level signatures. The binary AAD comes from collapsing 4-class labels into 2-class, which loses the lateralisation information the TRF is most sensitive to.

Subj	<i>n</i>	gaze	EEG (LogReg)	late	stacked	early
1	99	0.404	0.566	0.535	0.606	0.586
2	100	0.960	0.530	0.970	0.950	0.980
3	100	0.970	0.520	0.980	0.970	0.960
4	100	0.980	0.480	0.970	0.960	0.960
5	100	0.750	0.550	0.750	0.700	0.740
6	100	0.650	0.450	0.660	0.610	0.630
7	100	0.940	0.570	0.950	0.950	0.930
8	64	0.516	0.438	0.516	0.516	0.562
9	100	0.720	0.500	0.700	0.670	0.660
10	100	0.930	0.470	0.900	0.900	0.930
11	100	0.710	0.590	0.630	0.720	0.710
12	97	0.629	0.433	0.639	0.608	0.670
13	100	0.880	0.510	0.890	0.880	0.910
14	100	0.730	0.590	0.710	0.710	0.740
15	99	0.949	0.566	0.960	0.939	0.980
16	100	0.830	0.530	0.840	0.820	0.810

3. **EEG helps the subjects whose gaze is unreliable.** S1 goes from 0.40 (gaze) to 0.61 (stacked fusion); S12 from 0.63 to 0.67. These are the subjects who didn’t visually track the attended speaker.
4. **Early fusion > late fusion.** Learning a non-linear combination of raw features lets the model recover information that probability averaging throws away.
5. **Subject 1** has below-chance gaze *and* below-chance EEG — consistent with the known-bad mastoids + apparent inattention. Subject 8 has only 64/100 valid Tobii recordings (missing files).

6.3 Implications and iter-4 plan

6.3.1 What iter-3 settles

- The headline dataset finding is “**gaze-dominant multimodal AAD at 0.80 pooled (16 subjects, 30 s windows)**”.
- Linear EEG-TRF alone cannot drive group-level AAD on this paradigm. We need either a richer representation (deep model) or a different task framing (speaker-identity classification, speech-envelope reconstruction correlation).
- We have enough evidence to draft the paper’s results section and figures.

6.3.2 What to try in iter-4

1. **Deep EEG decoders** — EEGNet / DenseNet / cross-modal transformer on 5–10 s windows. Already implemented (RUN_DEEP=False); needs GPU or a long CPU run. Target: EEG-only ≥ 0.60 binary AAD.

2. **Gaze-excluded control** — can we decode attended direction from EEG alone when *gaze is regressed out* pre-analysis? This answers reviewer question “are you just doing eye-tracking AAD?”.
3. **Learning curves** — fusion accuracy vs. n_{trials} , to show the dataset’s sample efficiency.
4. **Subject-quality score** — combine bad-channel rate, α -lat. sign, gaze validity into one number; verify it predicts AAD accuracy.
5. **Training-trial transfer** — use the 5 training trials as per-subject domain-adaptation data.
6. **Deeper dive on S1, S13, S15** — the known-bad subjects need either exclusion or a dedicated failure-mode analysis.

6.3.3 Open questions to resolve before submission

- Do we keep all 16 subjects in the primary benchmark, or exclude the known-bad ones? (Dataset papers generally do not exclude; report both tiers and a quality-stratified analysis.)
- Is the paper a *dataset release* or a *method paper*? The iter-3 result (gaze-dominated fusion) is more dataset-release in flavour than a novel decoder contribution.